

ITSimplera



Assignment # 01

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1. Objective

The objective of this task is to perform Exploratory Data Analysis (EDA) on the Online Retail II dataset. The purpose is to understand customer purchasing behavior, identify data quality issues, analyze product sales, evaluate country-wise revenue, and generate business insights before implementing a recommendation system.

2. Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

3. Loading the Dataset

The Excel dataset was loaded using the Pandas library.

```
# Read the first sheet
df = pd.read_excel("online_retail_II.xlsx")
```

4. Dataset Information

The following information was analyzed:

- Number of Rows
- Number of Columns
- Data Types
- Dataset Information
- Statistical Summary

```
print("Rows:", df.shape[0])  
print("Columns:", df.shape[1])
```

✓ 0.0s

Python

```
Rows: 525461  
Columns: 8
```

```
print(df.columns)
```

✓ 0.0s

Python

```
Index(['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',  
      'Price', 'Customer ID', 'Country'],  
      dtype='object')
```

```
df.dtypes
```

✓ 0.0s

Python

```
Invoice          object  
StockCode        object  
Description       object  
Quantity         int64  
InvoiceDate      datetime64[ns]  
Price            float64  
Customer ID      float64  
Country          object  
dtype: object
```

```
df.info()
```

✓ 0.0s

Python

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525461 entries, 0 to 525460
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Invoice          525461 non-null object
1   StockCode       525461 non-null object
2   Description     522533 non-null object
3   Quantity        525461 non-null int64
4   InvoiceDate      525461 non-null datetime64[ns]
5   Price           525461 non-null float64
6   Customer ID     417534 non-null float64
7   Country         525461 non-null object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 32.1+ MB
```

```
df.describe()
```

✓ 0.1s

Python

	Quantity	InvoiceDate	Price	Customer ID
count	525461.000000	525461	525461.000000	417534.000000
mean	10.337667	2010-06-28 11:37:36.845017856	4.688834	15360.645478
min	-9600.000000	2009-12-01 07:45:00	-53594.360000	12346.000000
25%	1.000000	2010-03-21 12:20:00	1.250000	13983.000000
50%	3.000000	2010-07-06 09:51:00	2.100000	15311.000000
75%	10.000000	2010-10-15 12:45:00	4.210000	16799.000000
max	19152.000000	2010-12-09 20:01:00	25111.090000	18287.000000
std	107.424110	NaN	146.126914	1680.811316

5. Missing Values Analysis

Missing values were identified to understand the quality of the dataset.

```
missing = df.isnull().sum()

print(missing)
```

```
Invoice      0
StockCode    0
Description  2928
Quantity     0
InvoiceDate  0
Price        0
Customer ID  107927
Country      0
dtype: int64
```

6. Duplicate Rows

```
duplicates = df.duplicated().sum()

print("Duplicate Rows =", duplicates)
```

✓ 0.2s

```
Duplicate Rows = 6865
```

7. Revenue Calculation

A new Revenue column was created using the following formula:

$$\text{Revenue} = \text{Quantity} \times \text{Price}$$

```
df["Revenue"] = df["Quantity"] * df["Price"]  
  
df.head()  
✓ 0.0s Python
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country	Revenue
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom	83.4
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom	81.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom	81.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom	100.8
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom	30.0

8. Top Selling Products

1.Top Product by Quantity Sold

```
top_quantity = (  
    df.groupby("Description")["Quantity"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
)
```

```
print(top_quantity)
```

✓ 0.0s

Python

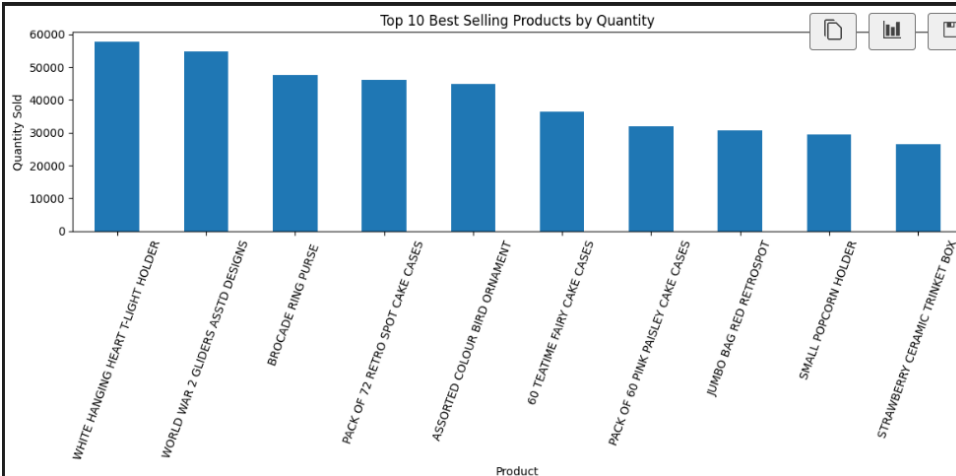
Description	
WHITE HANGING HEART T-LIGHT HOLDER	57733
WORLD WAR 2 GLIDERS ASSTD DESIGNS	54698
BROCADE RING PURSE	47647
PACK OF 72 RETRO SPOT CAKE CASES	46106
ASSORTED COLOUR BIRD ORNAMENT	44925
60 TEATIME FAIRY CAKE CASES	36326
PACK OF 60 PINK PAISLEY CAKE CASES	31822
JUMBO BAG RED RETROSPOT	30727
SMALL POPCORN HOLDER	29500
STRAWBERRY CERAMIC TRINKET BOX	26563

Name: Quantity, dtype: int64

```
plt.figure(figsize=(12,6))  
  
top_quantity.plot(kind="bar")  
  
plt.title("Top 10 Best Selling Products by Quantity")  
  
plt.xlabel("Product")  
  
plt.ylabel("Quantity Sold")  
  
plt.xticks(rotation=70)  
  
plt.tight_layout()  
  
plt.show()
```

✓ 0.4s

Python



2. Top Product by Revenue

```
top_revenue = (  
    df.groupby("Description")["Revenue"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
)
```

```
print(top_revenue)
```

✓ 0.0s

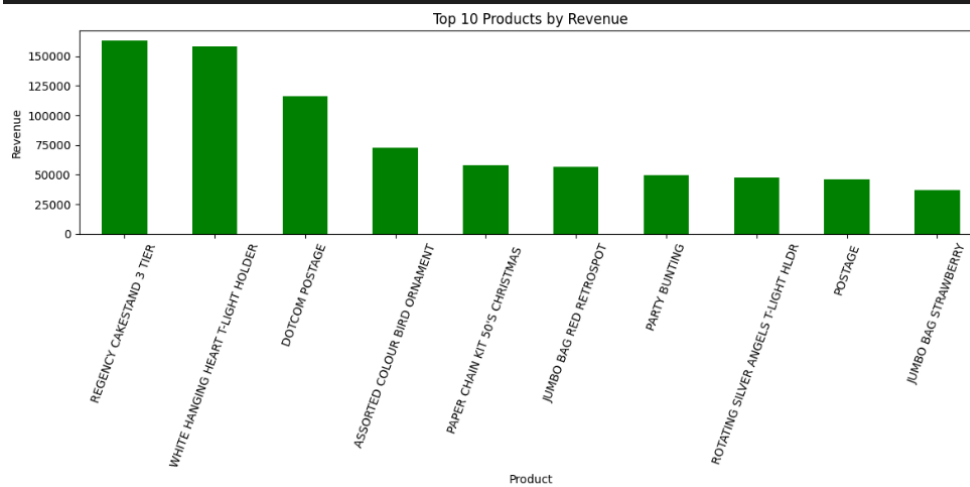
Python

```
Description  
REGENCY CAKESTAND 3 TIER      163051.46  
WHITE HANGING HEART T-LIGHT HOLDER  157865.43  
DOTCOM POSTAGE                116401.99  
ASSORTED COLOUR BIRD ORNAMENT    72454.12  
PAPER CHAIN KIT 50'S CHRISTMAS   57870.20  
JUMBO BAG RED RETROSPOT         56444.61  
PARTY BUNTING                 49645.52  
ROTATING SILVER ANGELS T-LIGHT HLDR  47672.49  
POSTAGE                        46092.36  
JUMBO BAG STRAWBERRY           36699.61  
Name: Revenue, dtype: float64
```

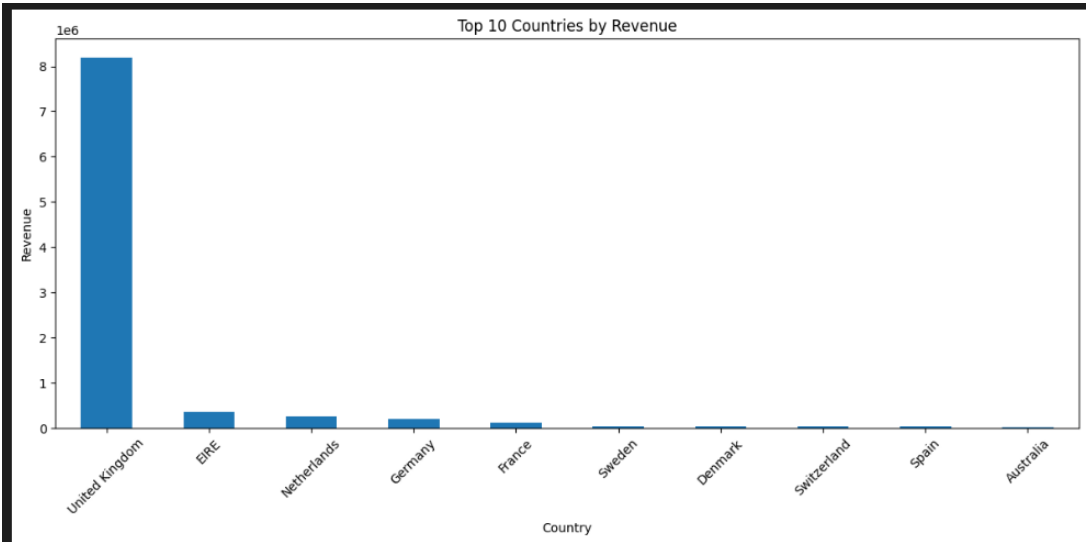
```
plt.figure(figsize=(12,6))  
  
top_revenue.plot(kind="bar",color="green")  
  
plt.title("Top 10 Products by Revenue")  
  
plt.xlabel("Product")  
  
plt.ylabel("Revenue")  
  
plt.xticks(rotation=70)  
  
plt.tight_layout()  
  
plt.show()
```

✓ 0.3s

Python



9. Sales Performance by Country



10. Monthly Revenue Trend

```
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])
```

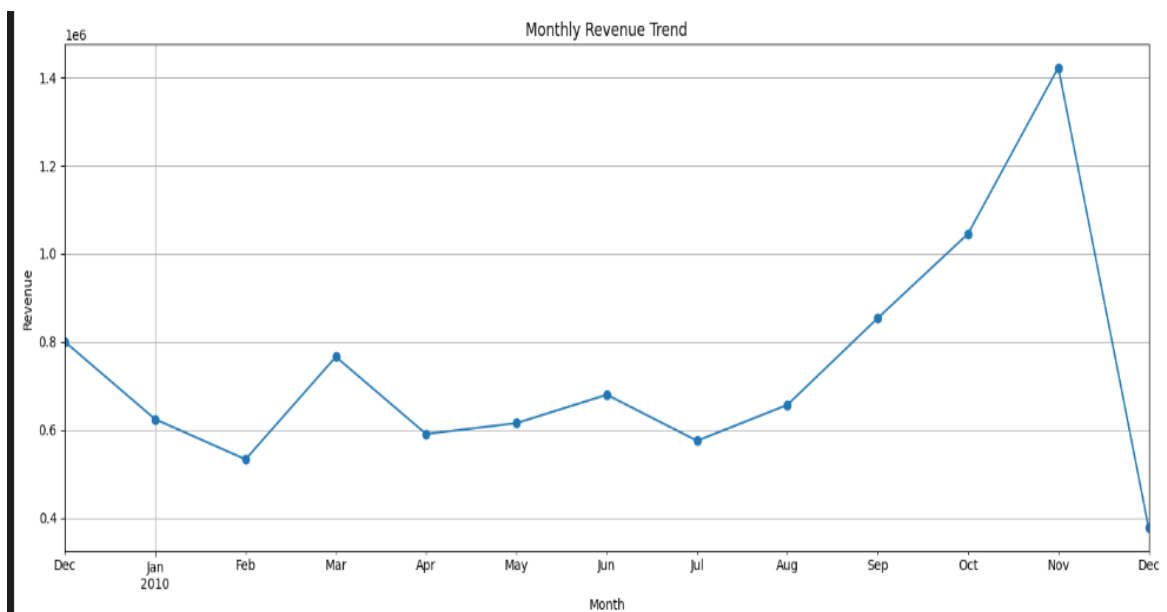
```
df["Month"] = df["InvoiceDate"].dt.to_period("M")
```

```
monthly_sales = (  
    df.groupby("Month")["Revenue"]  
    .sum()  
)
```

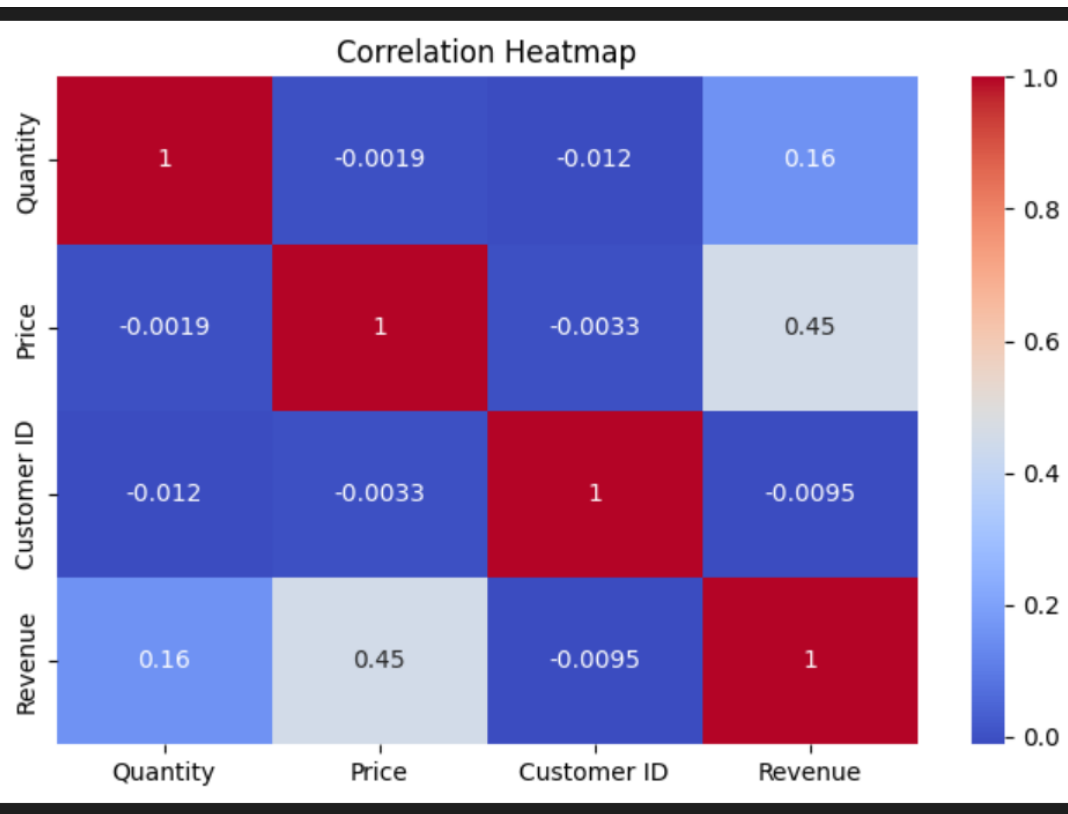
```
monthly_sales
```

✓ 0.0s

Month	
2009-12	799847.110
2010-01	624032.892
2010-02	533091.426
2010-03	765848.761
2010-04	590580.432
2010-05	615322.830
2010-06	679786.610
2010-07	575236.360
2010-08	656776.340
2010-09	853650.431
2010-10	1045168.350
2010-11	1422654.642
2010-12	377488.450

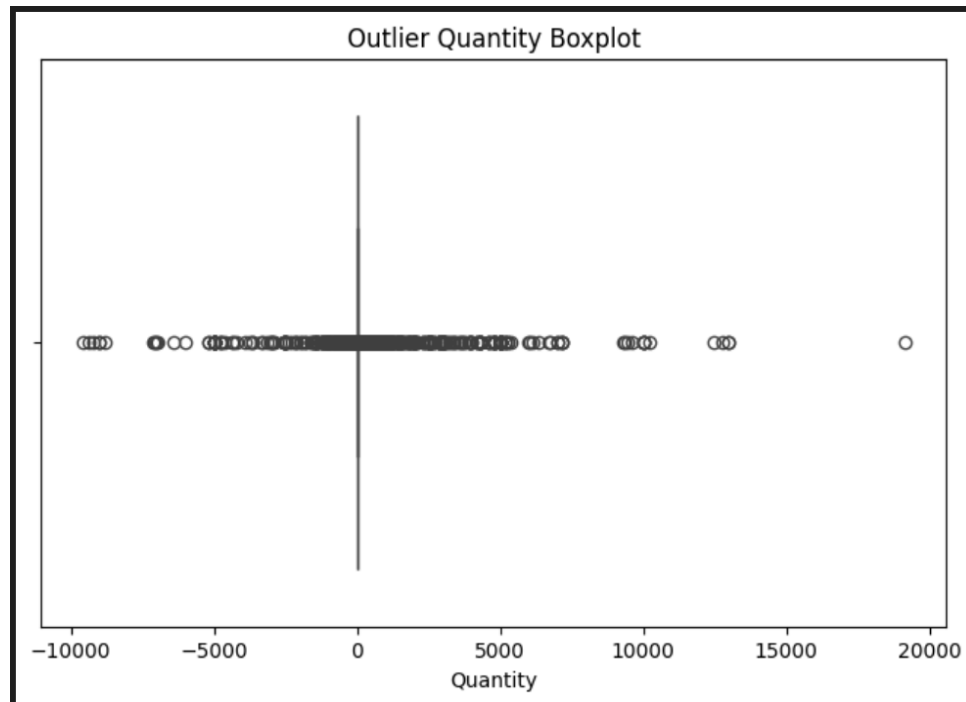


11. Correlation Heatmap

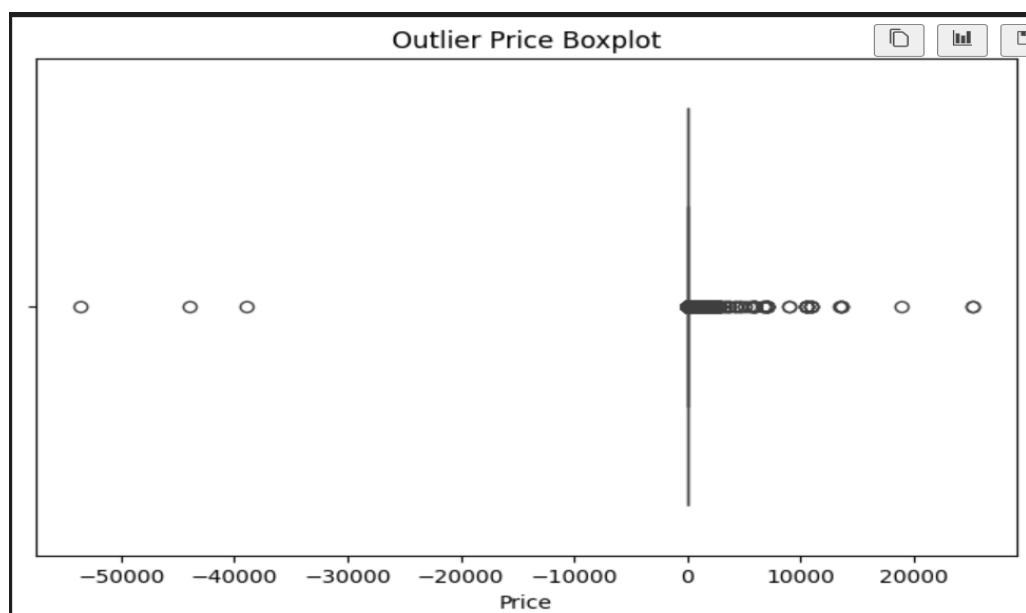


12. Outlier Detection

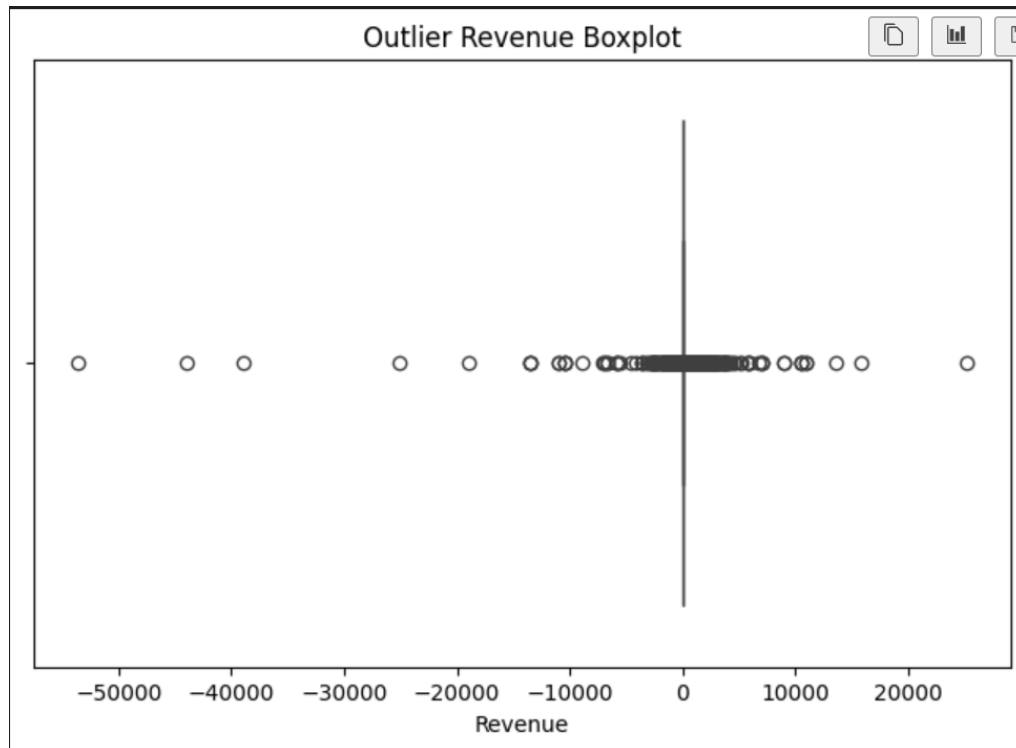
1.Quantity Outlier



2.Price Outlier



3.Revenue Outlier



13. Business Insights

Insight 1

The United Kingdom contributes the highest revenue among all countries.

Insight 2

A small number of products generate a large percentage of overall sales.

Insight 3

The dataset contains missing Customer IDs and Product Descriptions that require cleaning in future stages.

Insight 4

Revenue changes significantly across different months, indicating seasonal customer purchasing behavior.

Insight 5

Several extreme outliers exist in Quantity, Price, and Revenue, suggesting bulk purchases, returns, or unusual transactions.

Insight 6

Products with higher sales quantities generally contribute more to total revenue.

14. Conclusion

The Exploratory Data Analysis successfully provided an overview of customer purchasing behavior and sales performance. The analysis identified data quality issues, high-performing products, country-wise revenue distribution, monthly revenue trends, and potential outliers. These findings provide valuable insights that can be used in future stages for data cleaning, feature engineering, and recommendation system development.